

# Image editing and customization with diffusion models

Customizing diffusion models using  
example images

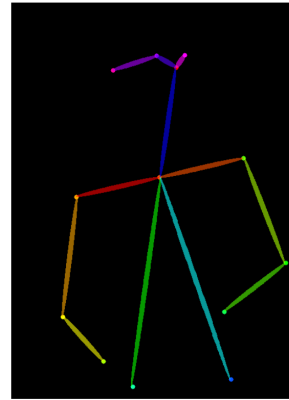


Input images



*in the Acropolis*

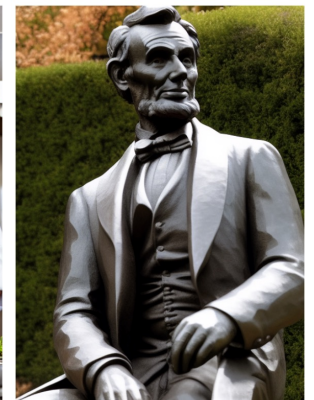
Controlling diffusion models using pixel-aligned conditions



Input human pose



“Chef in kitchen”



“Lincoln statue”

Text-driven image editing



Input



“Make it a Modigliani painting”

# Customizing DMs: Textual inversion

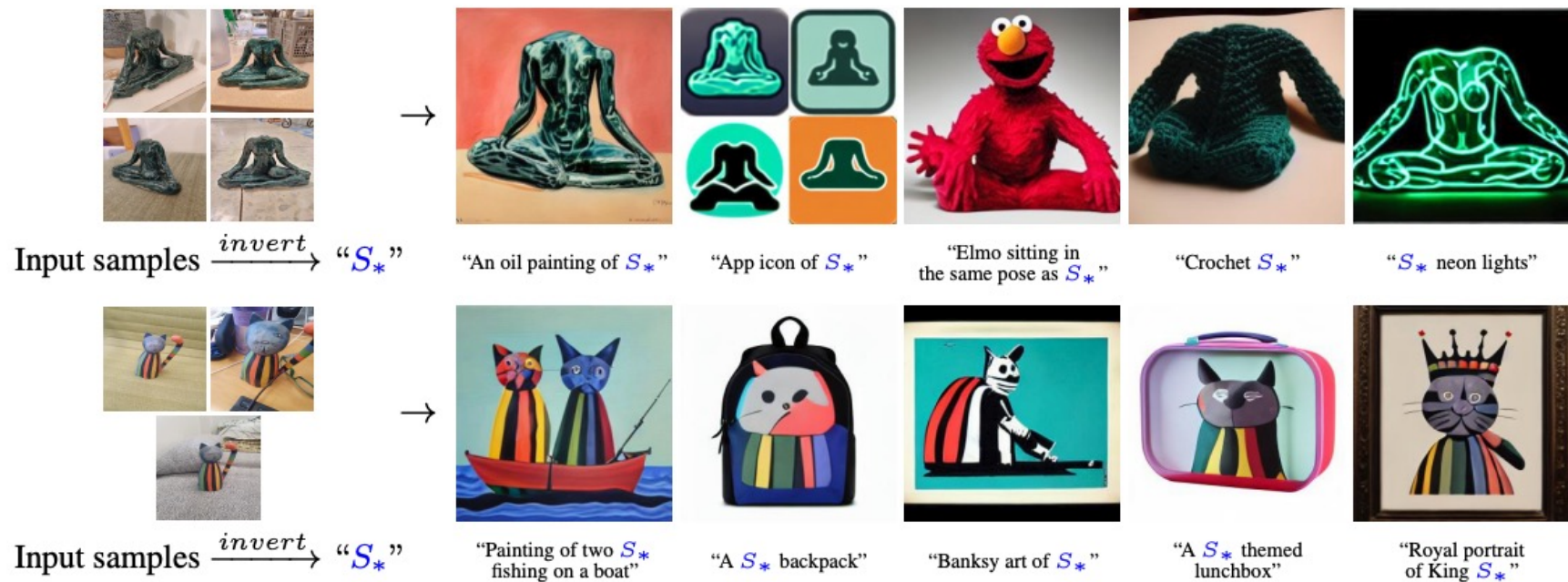


Figure 1: (left) We find new pseudo-words in the embedding space of pre-trained text-to-image models which describe specific concepts. (right) These pseudo-words are composed into new sentences, placing our targets in new scenes, changing their style or ingraining them into new products.

# Customizing DMs: Textual inversion

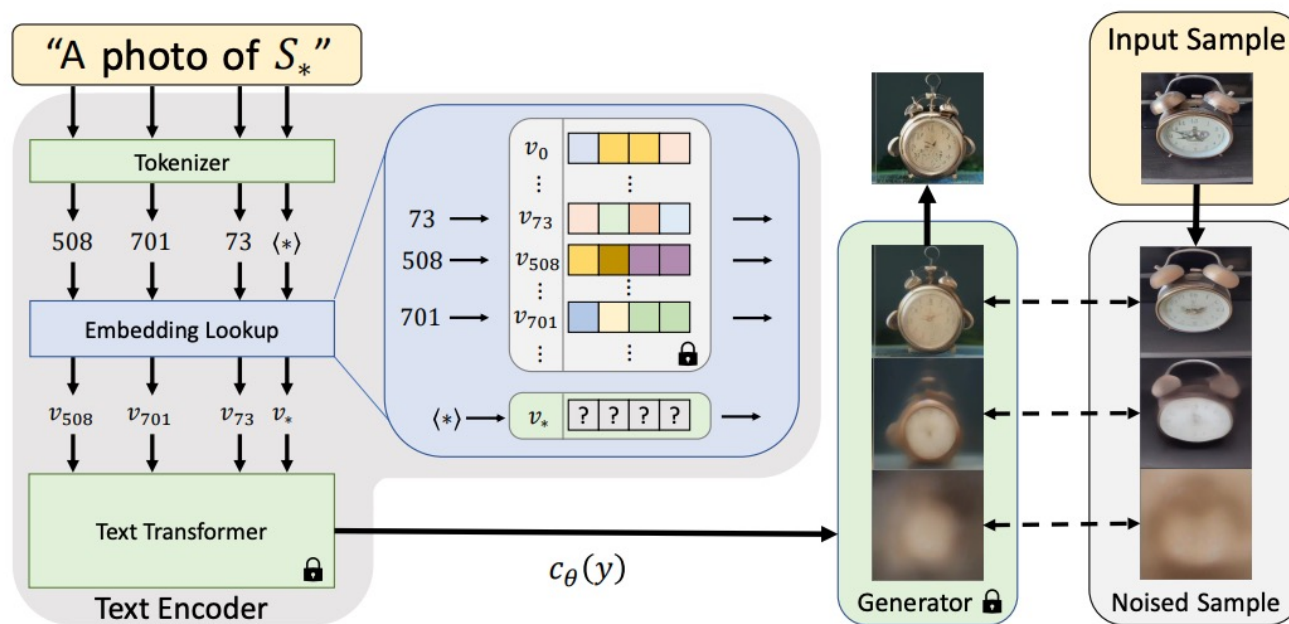


Figure 2: Outline of the text-embedding and inversion process. A string containing our placeholder word is first converted into tokens (*i.e.* word or sub-word indices in a dictionary). These tokens are converted to continuous vector representations (the “embeddings”,  $v$ ). Finally, the embedding vectors are transformed into a conditioning code  $c_\theta(y)$  that guides the generation. We optimize the embedding vector  $v_*$  associated with our pseudo-word  $S_*$ , using a reconstruction objective.



# Textual inversion: Results



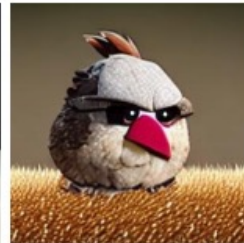
Input samples



"Watercolor painting of  $S_*$  on a branch"



"A house in the style of  $S_*$ "



"Grainy photo of  $S_*$  in angry birds"



" $S_*$  made of chocolate"



"A  $S_*$  dragon"



Input samples



"A mosaic depicting  $S_*$ "



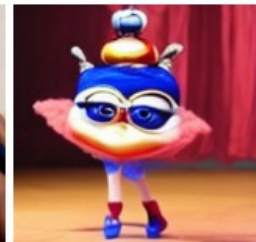
"Death metal album cover featuring  $S_*$ "



"Masterful oil painting of  $S_*$  hanging on the wall"



"An artist drawing a  $S_*$ "



"A  $S_*$  dancing ballet"



Input samples



"A photo of  $S_*$  full of cashew nuts"



"A mouse using  $S_*$  as a boat"



"A photo of a  $S_*$  mask"



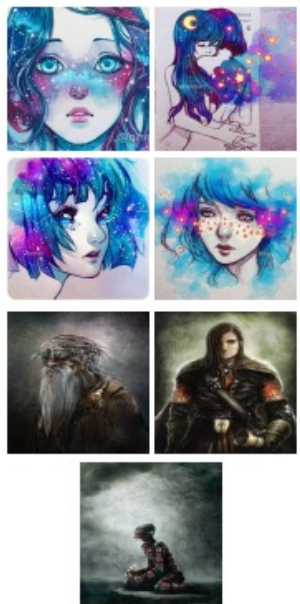
"Ramen soup served in  $S_*$ "



"Cave mural depicting  $S_*$ "



# Textual inversion: Results



Input samples



"The streets of Paris  
in the style of  $S_*$ "



"Adorable corgi  
in the style of  $S_*$ "



"Painting of a black hole  
in the style of  $S_*$ "



"Times square  
in the style of  $S_*$ "



"Edo period pagoda  
in the style of  $S_*$ "



# Textual inversion: Comparisons

|                         | Skull Mug                                                                                                                                                               | Teapot                                                                                                                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input samples           |                                                                                       |                                                                                               |
| Ours                    | <br>"A photo of $S_*$ "                                                               | <br>"A photo of $S_*$ "                                                                       |
| DALLE-2 (Image Inputs)  |                                                                                       |                                                                                               |
| DALLE-2 (Long Captions) |                                                                                     |                                                                                             |
| LDM (Long Captions)     | <br>A mug having many skulls at the bottom and sculpture of a man at the top of it. | <br>A tea pot, with green, red, blue, yellow, an apple with leaves and a lemon with leaves. |

# Customizing DMs: DreamBooth

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Input images



in the Acropolis



swimming



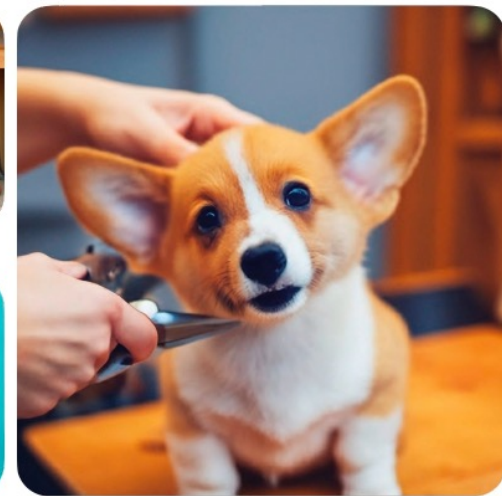
sleeping



in a doghouse



in a bucket



getting a haircut

N. Ruiz et al. [DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation](#). CVPR 2023



# Customizing DMs: DreamBooth

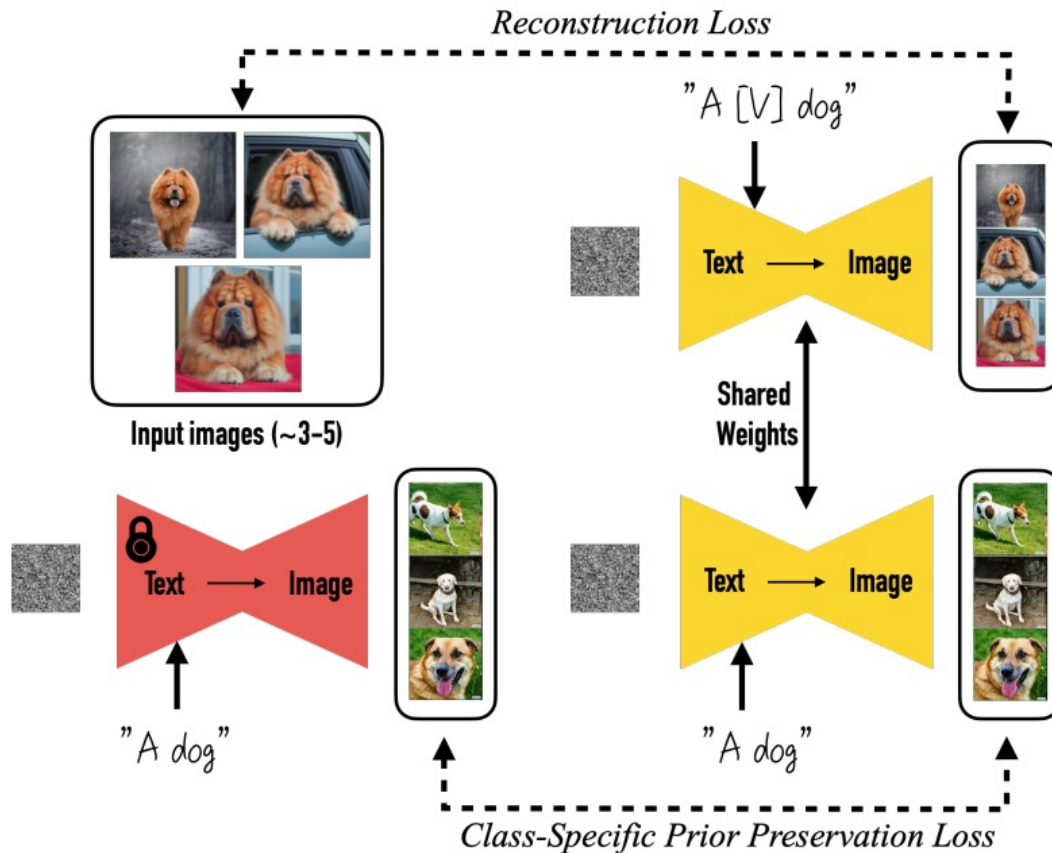


Figure 3. **Fine-tuning.** Given  $\sim 3 - 5$  images of a subject we fine-tune a text-to-image diffusion model with the input images paired with a text prompt containing a unique identifier and the name of the class the subject belongs to (e.g., "A [V] dog"), in parallel, we apply a class-specific prior preservation loss, which leverages the semantic prior that the model has on the class and encourages it to generate diverse instances belong to the subject's class using the class name in a text prompt (e.g., "A dog").

# Customizing DMs: DreamBooth

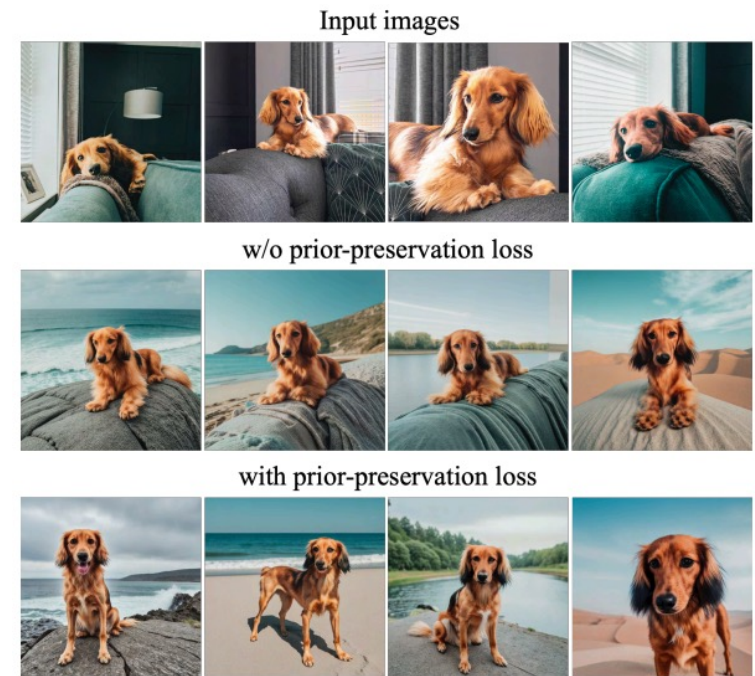
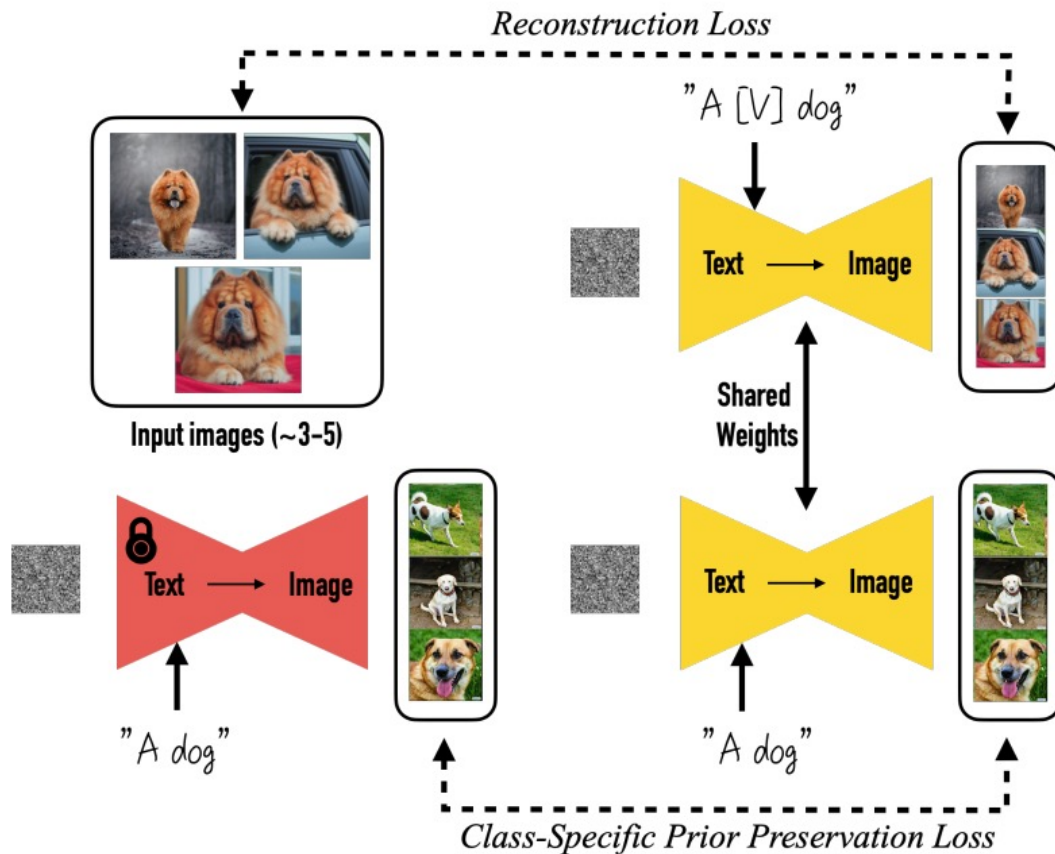


Figure 6. **Encouraging diversity with prior-preservation loss.** Naive fine-tuning can result in overfitting to input image context and subject appearance (e.g. pose). PPL acts as a regularizer that alleviates overfitting and encourages diversity, allowing for more pose variability and appearance diversity.

N. Ruiz et al. [DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation](#). CVPR 2023

# DreamBooth: Comparison with textual inversion

Input Images



DreamBooth (Imagen)



DreamBooth (Stable Diffusion)



Textual Inversion (Stable Diffusion)



"a [V] vase in  
the snow"

"a [V] vase on  
the beach"

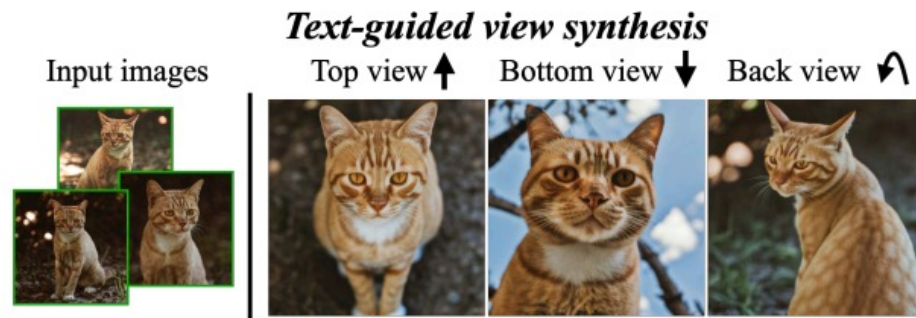
"a [V] vase in  
the jungle"

"a [V] vase with Eiffel Tower in  
the background"

| Method                               | Subject Fidelity ↑ | Prompt Fidelity ↑ |
|--------------------------------------|--------------------|-------------------|
| DreamBooth (Stable Diffusion)        | <b>68%</b>         | <b>81%</b>        |
| Textual Inversion (Stable Diffusion) | 22%                | 12%               |
| Undecided                            | 10%                | 7%                |



# DreamBooth: Results



“a [painting/sculpture] of a [V] [class noun] in the style of [famous artist]”



“a cross of a [V] dog and a [target species]”

# DreamBooth: Results

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Input images



A [V] backpack in the Grand Canyon



A wet [V] backpack in water



A [V] backpack in Boston



A [V] backpack with the night sky



Input images



A [V] teapot floating in milk



A transparent [V] teapot with milk inside



A [V] teapot pouring tea



A [V] teapot floating in the sea

# DreamBooth: Limitations

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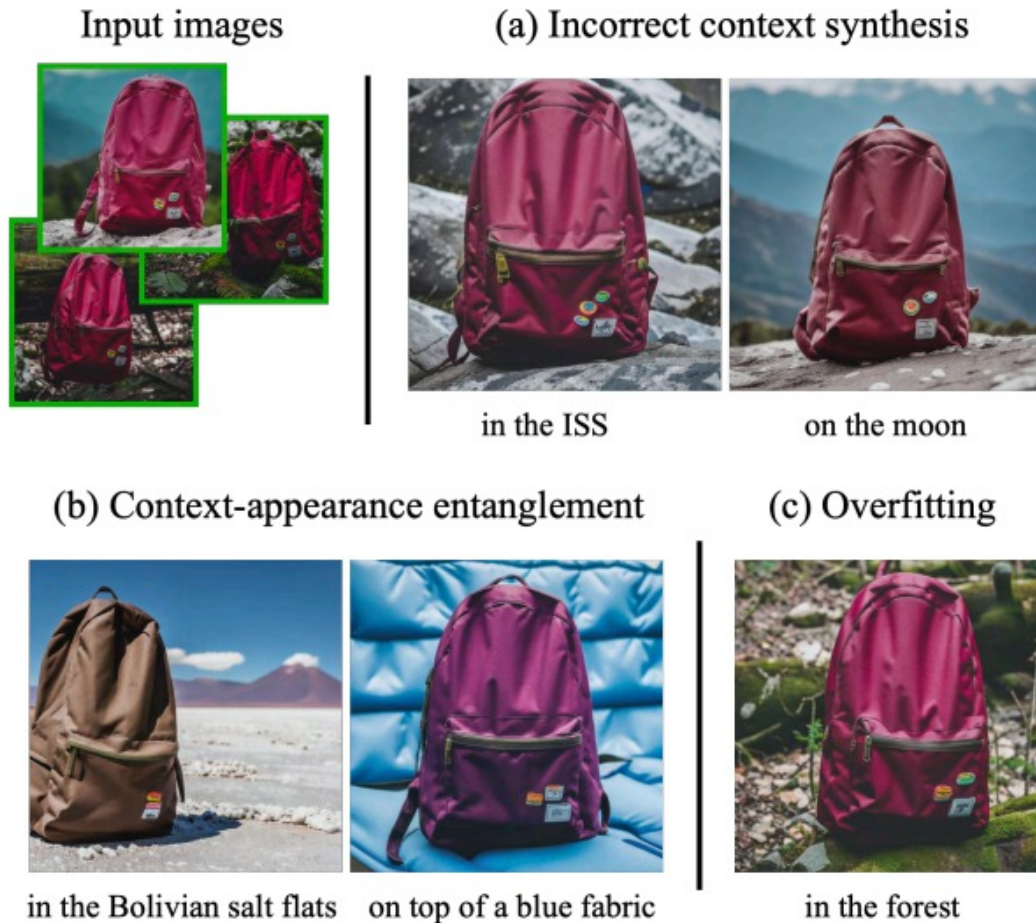
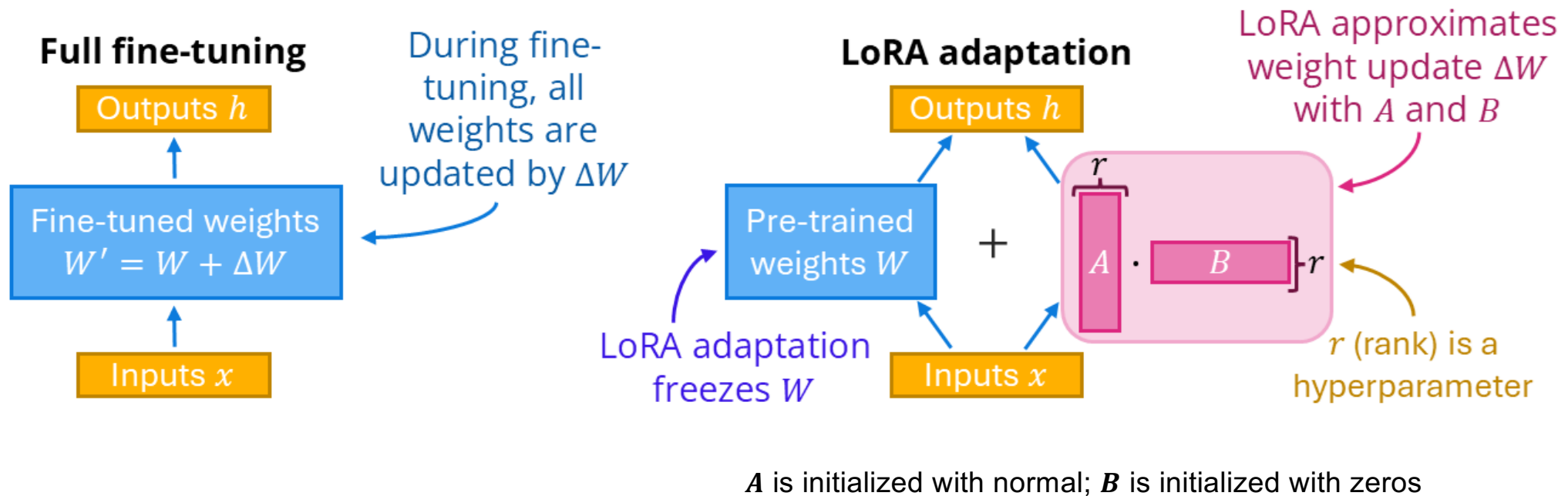


Figure 9. **Failure modes.** Given a rare prompted context the model might fail at generating the correct environment (a). It is possible for context and subject appearance to become entangled (b). Finally, it is possible for the model to overfit and generate images similar to the training set, especially if prompts reflect the original environment of the training set (c).



# Efficient Customization: DreamBooth with LoRA



[Figure source](#)

E. Hu et al. [LoRA: Low-Rank Adaptation of Large Language Models](#). ICLR'22

# LoRA DreamBooth: Results

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Input Images

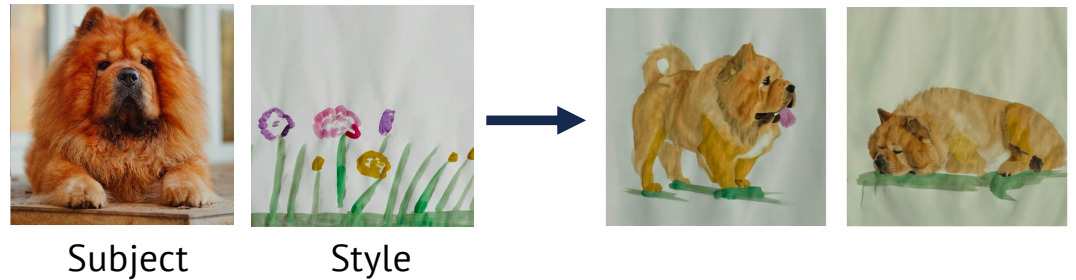


LoRA DreamBooth (r=4)



DreamBooth





# ZipLoRA: Any Subject in Any Style by Effectively Merging LoRAs

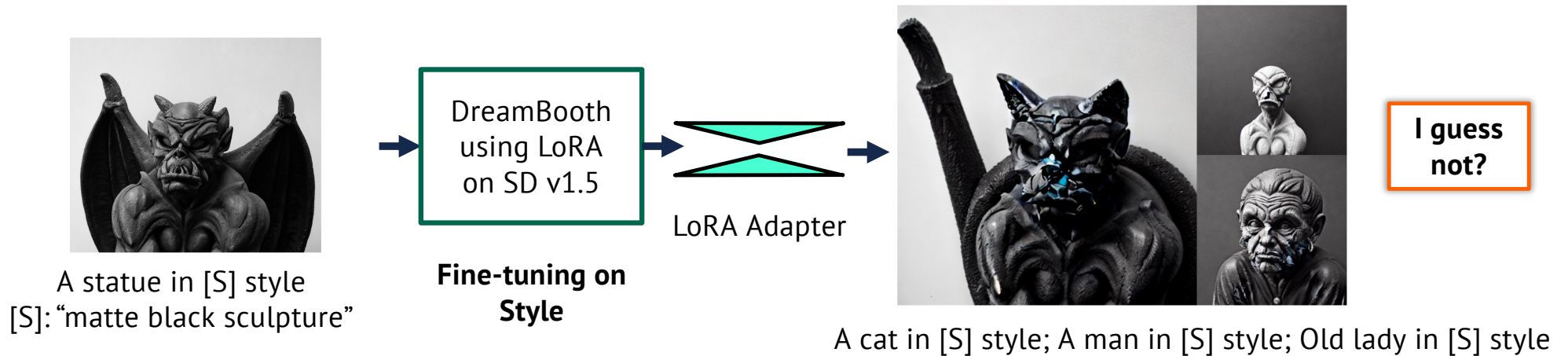
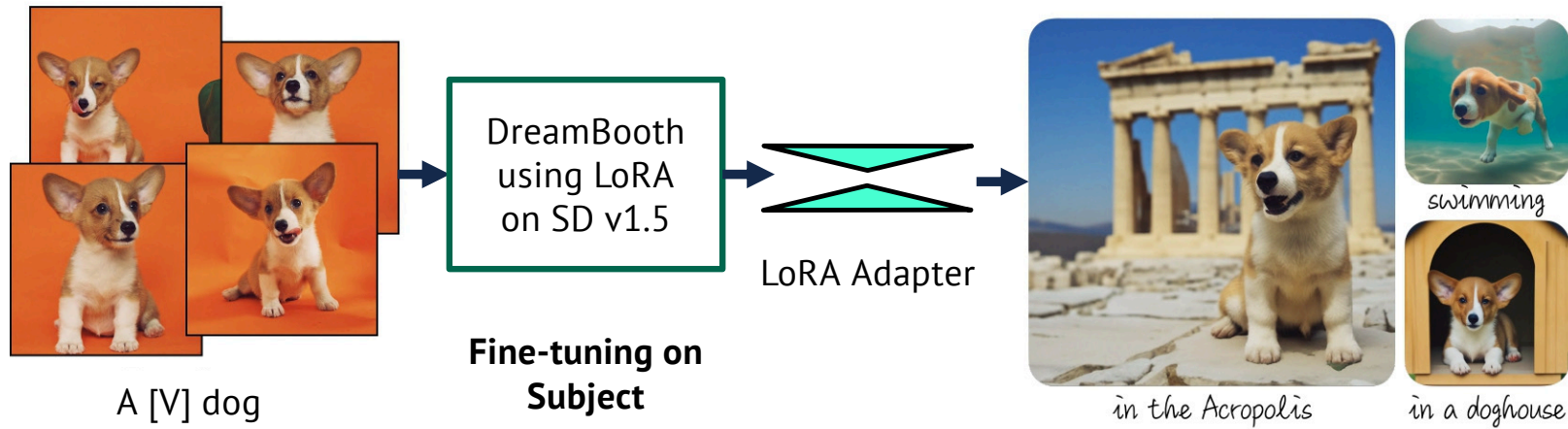
Viraj Shah, Nataniel Ruiz, Forrester Cole, Erika Lu,  
Svetlana Lazebnik, Yuanzhen Li, Varun Jampani

ECCV 2024

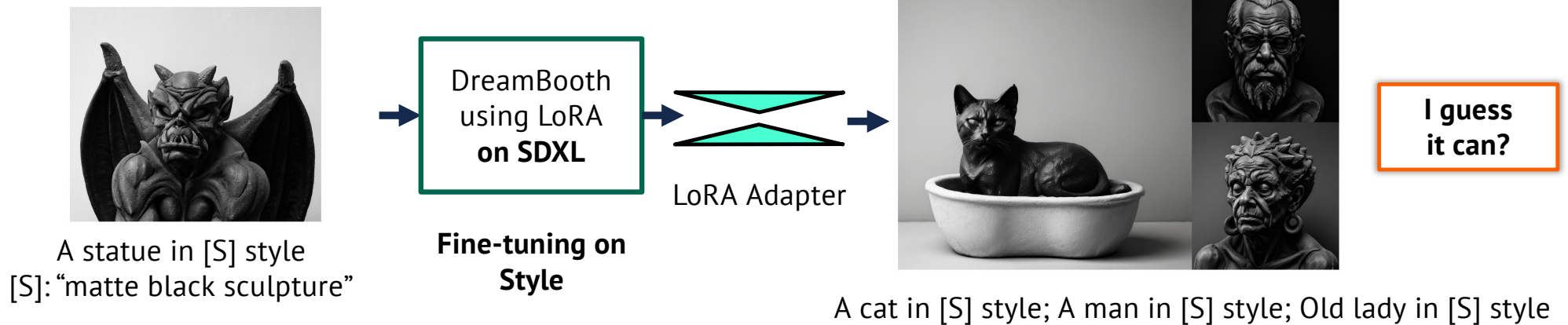
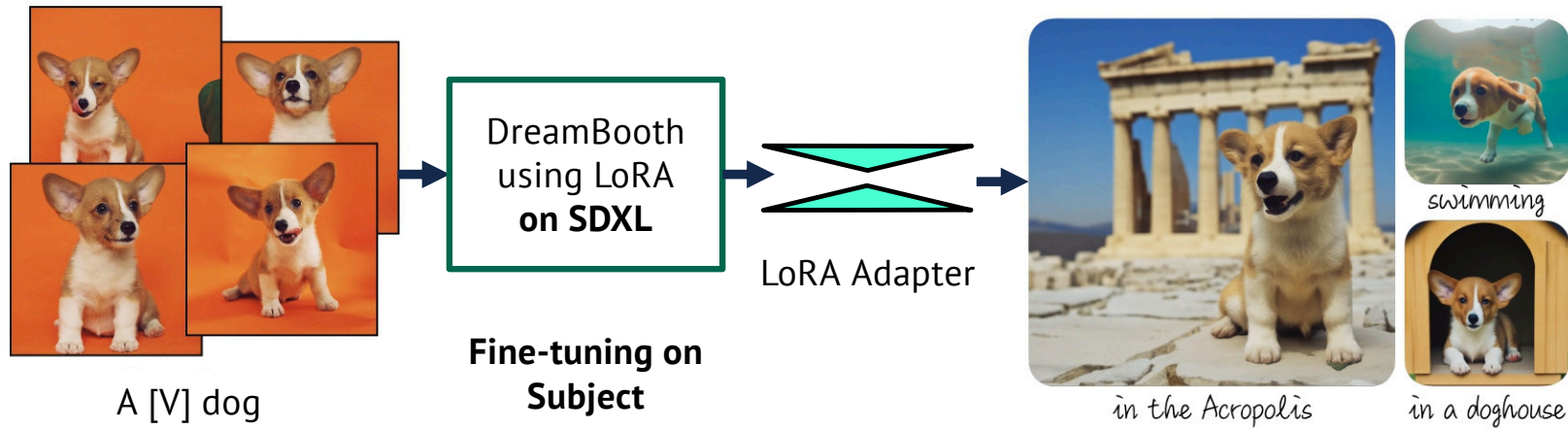




# Can DreamBooth learn a style?



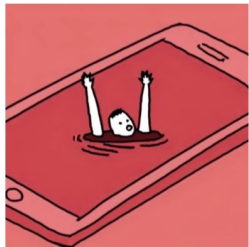
# Can DreamBooth learn a style?





# DreamBooth LoRA on SDXL for stylization

Style Reference



cartoon line drawing

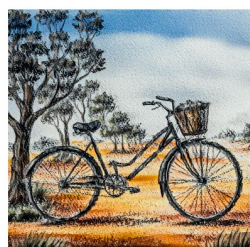


watercolor painting

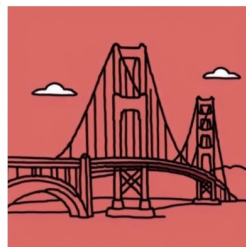


watercolor painting

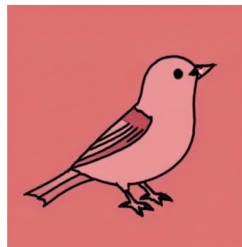
A bicycle in [S]  
Style



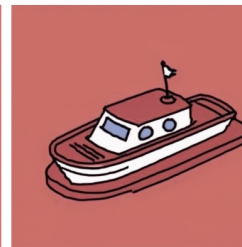
Golden gate  
bridge in [S] Style



A bird in  
[S] Style



A boat in  
[S] Style



A hat in  
[S] Style



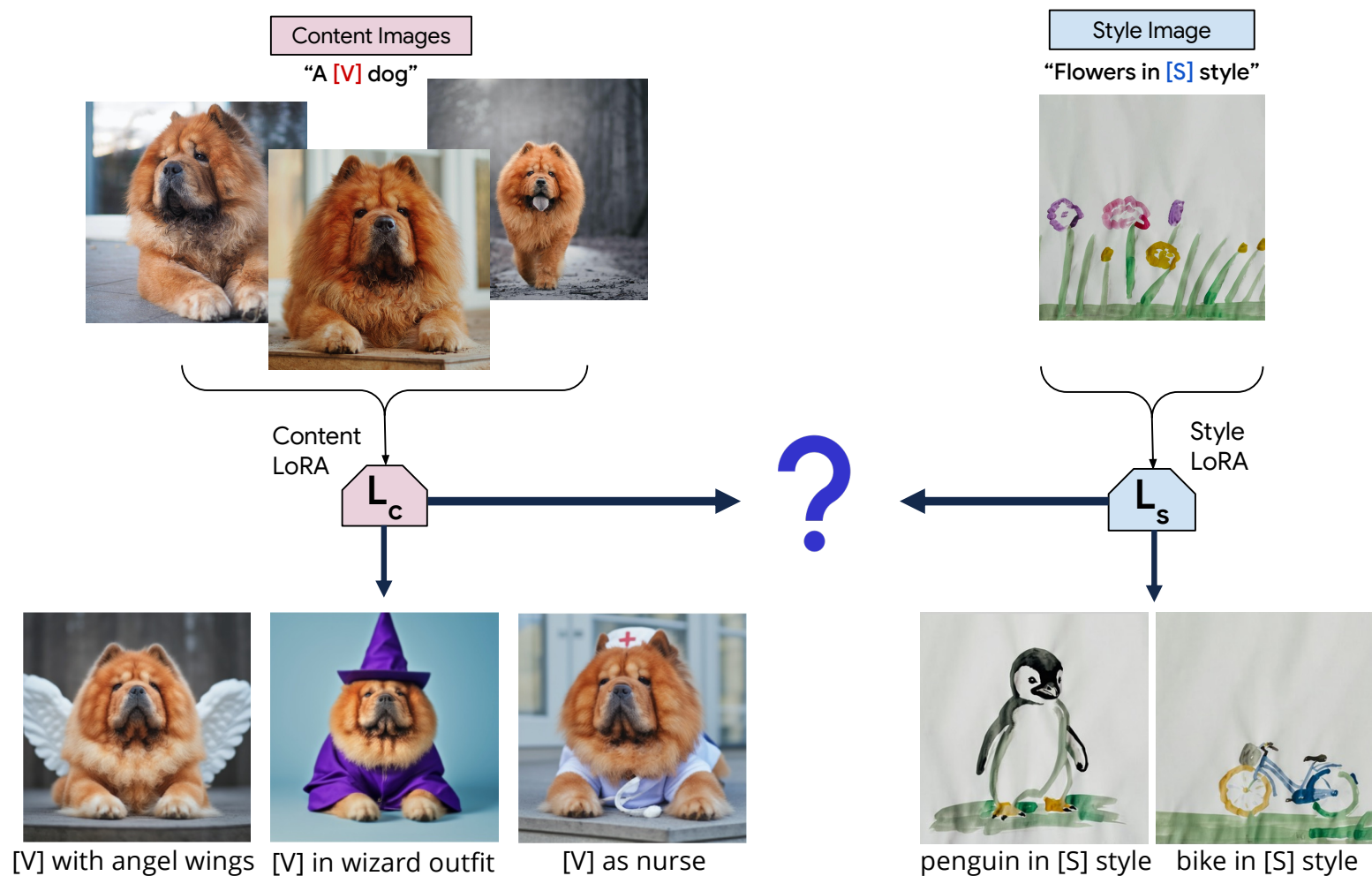
A piano in  
[S] Style



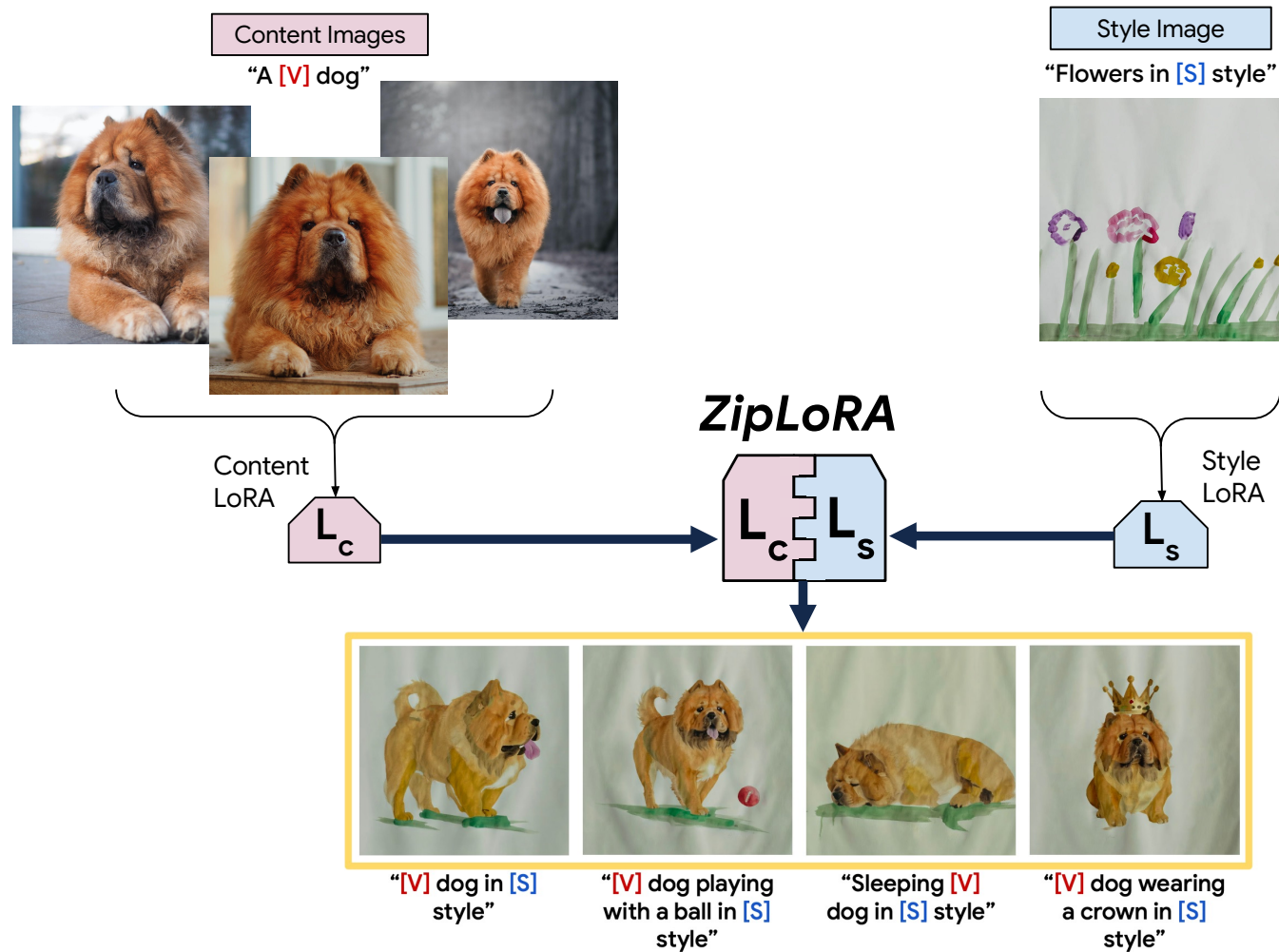
**Stylizations obtained using DreamBooth on SDXL with LoRA**



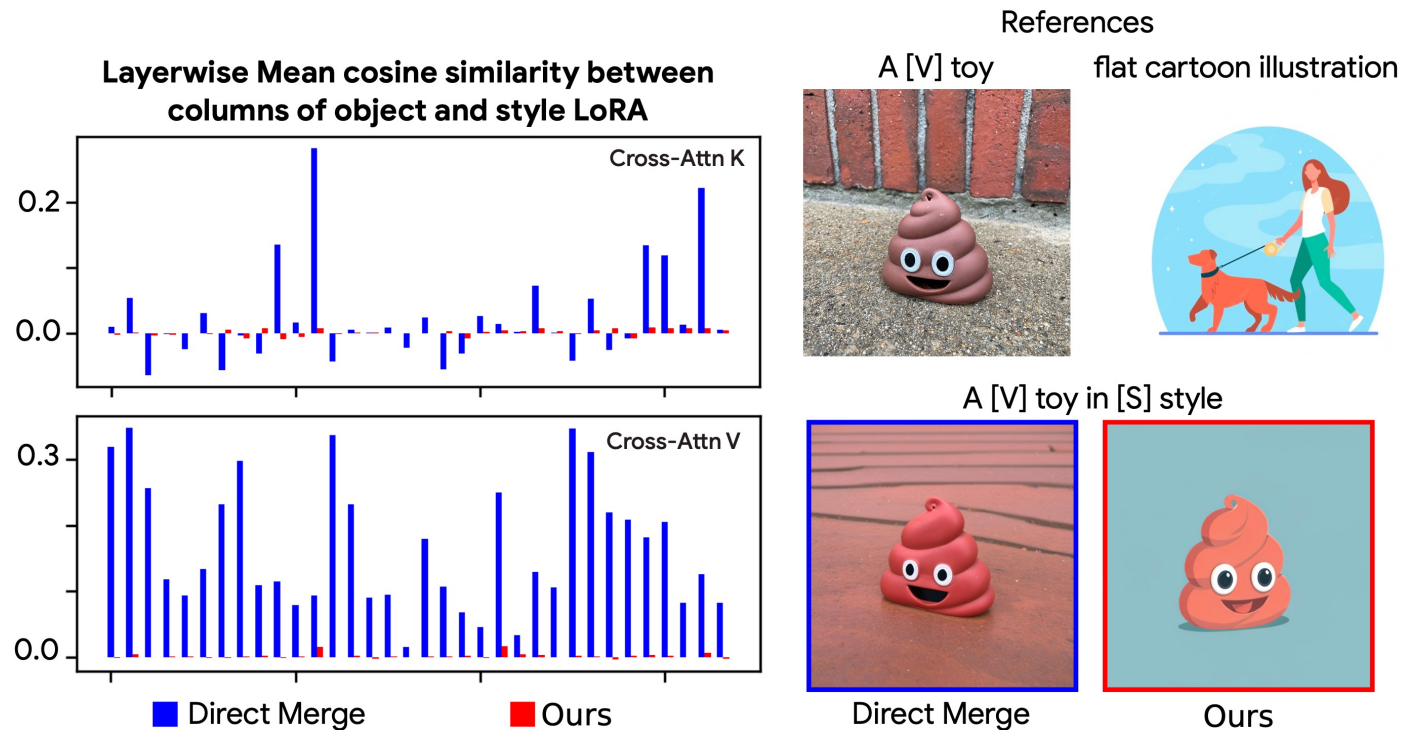
# Can we Merge Content and Style LoRAs?



# Can we Merge Content and Style LoRAs?



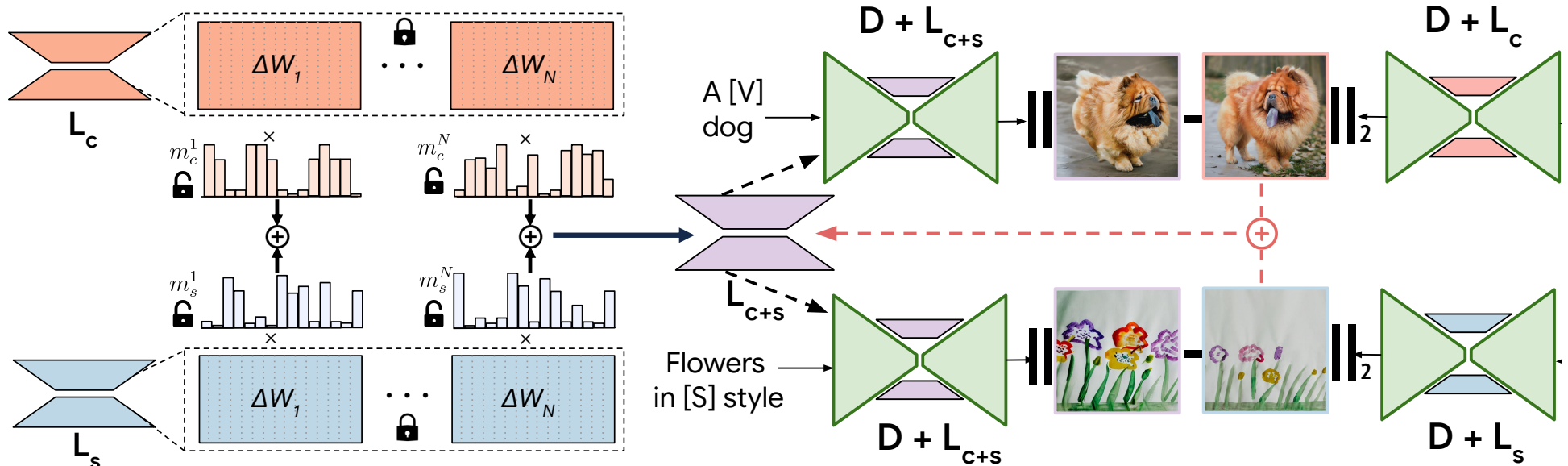
# Properties of LoRA weights: Insight



**Orthogonality matters!**



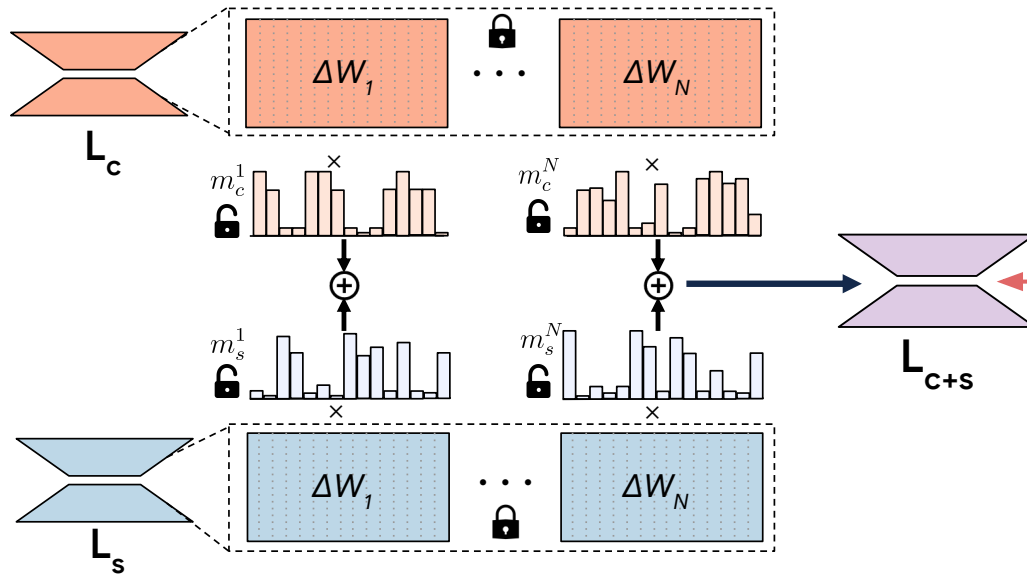
# ZipLoRA: Method



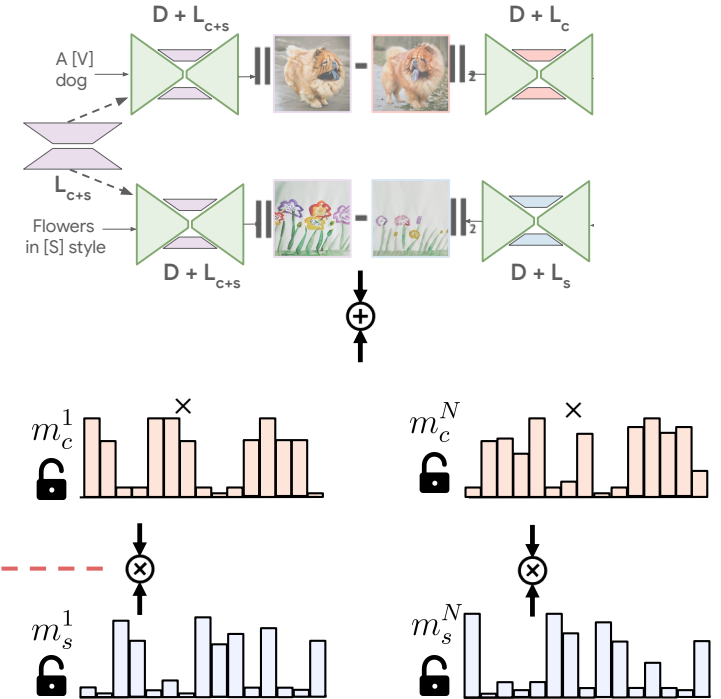
Learn mixing coefficients  $m_c^i$  and  $m_s^i$  for each column of  $\Delta W_i$  for both content and style LoRAs

Minimize the difference between the images generated by the mixed and original LoRAs

# ZipLoRA: Method



Learn mixing coefficients  $m_c^i$  and  $m_s^i$  for each column of  $\Delta W_i$  for both content and style LoRAs



Alignment loss: Minimize the cosine similarity between the coefficient vectors of the content and style LoRAs

A [V] toy in



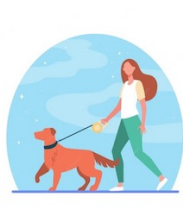
watercolor painting style



kid line drawing style



flat cartoon illustration style



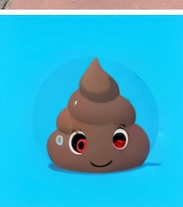
Direct arithmetic merge



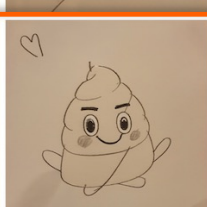
Joint Training



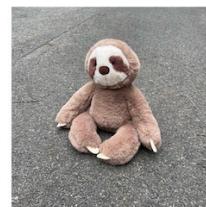
StyleDrop



Ours



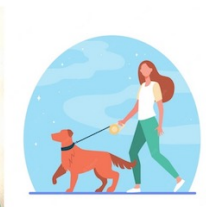
A [V] stuffed animal in



watercolor painting style



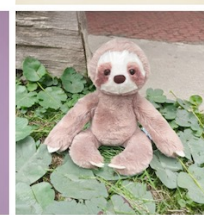
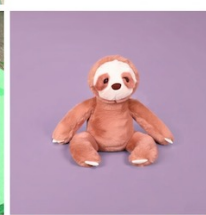
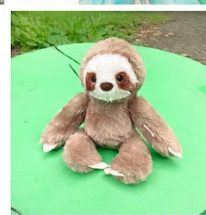
flat cartoon illustration style



watercolor painting style



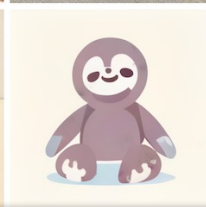
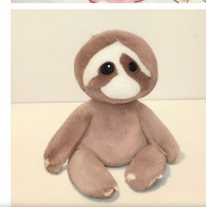
Direct arithmetic merge



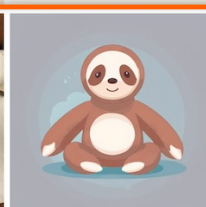
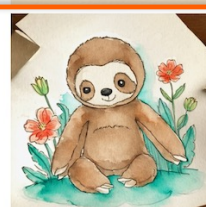
Joint Training



StyleDrop



Ours

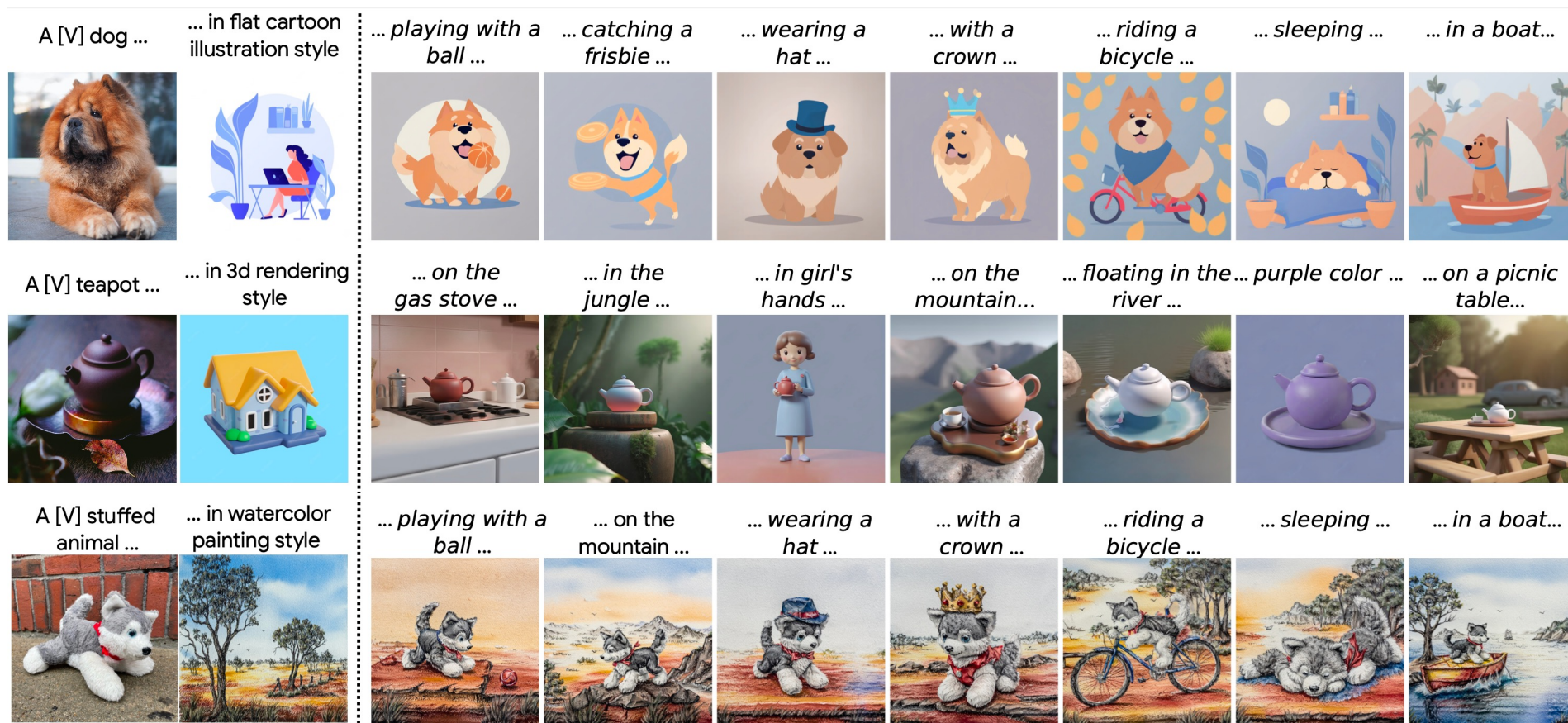




# ZipLoRA produces successful recontextualizations

Subject and style references

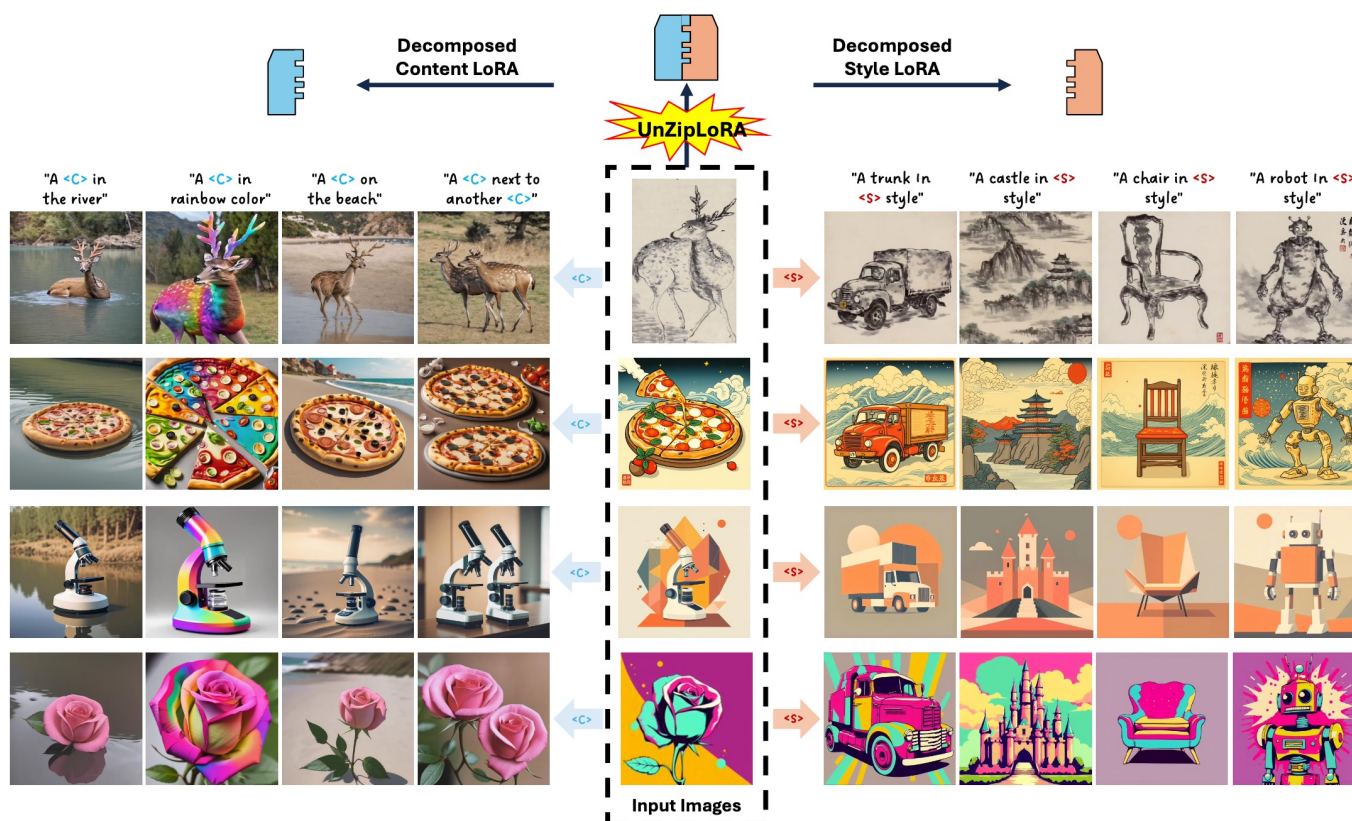
Recontextualizations using our method



# UnZipLoRA: Separating Content and Style from a Single Image

Chang Liu, Viraj Shah\*, Aiyu Cui†, Svetlana Lazebnik  
University of Illinois, Urbana-Champaign

changl25, vjshah3, aiyucui2, slazebni@illinois.edu



# Outline

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Customizing diffusion models using  
example images

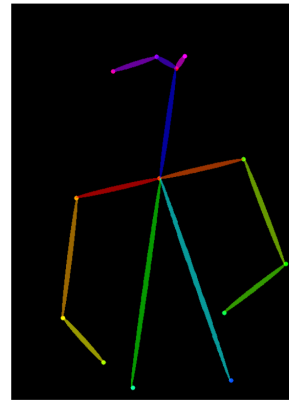


Input images



*in the Acropolis*

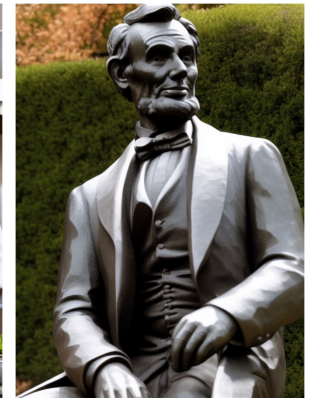
Controlling diffusion models using pixel-aligned conditions



Input human pose



“Chef in kitchen”



“Lincoln statue”



# ControlNet

- Add a trainable “wrapper” around a pre-trained DM to get it to follow pixel-aligned conditions



L. Zhang et al. [Adding Conditional Control to Text-to-Image Diffusion Models](#). ICCV 2023

# ControlNet

- Add a trainable “wrapper” around a pre-trained DM to get it to follow pixel-aligned conditions

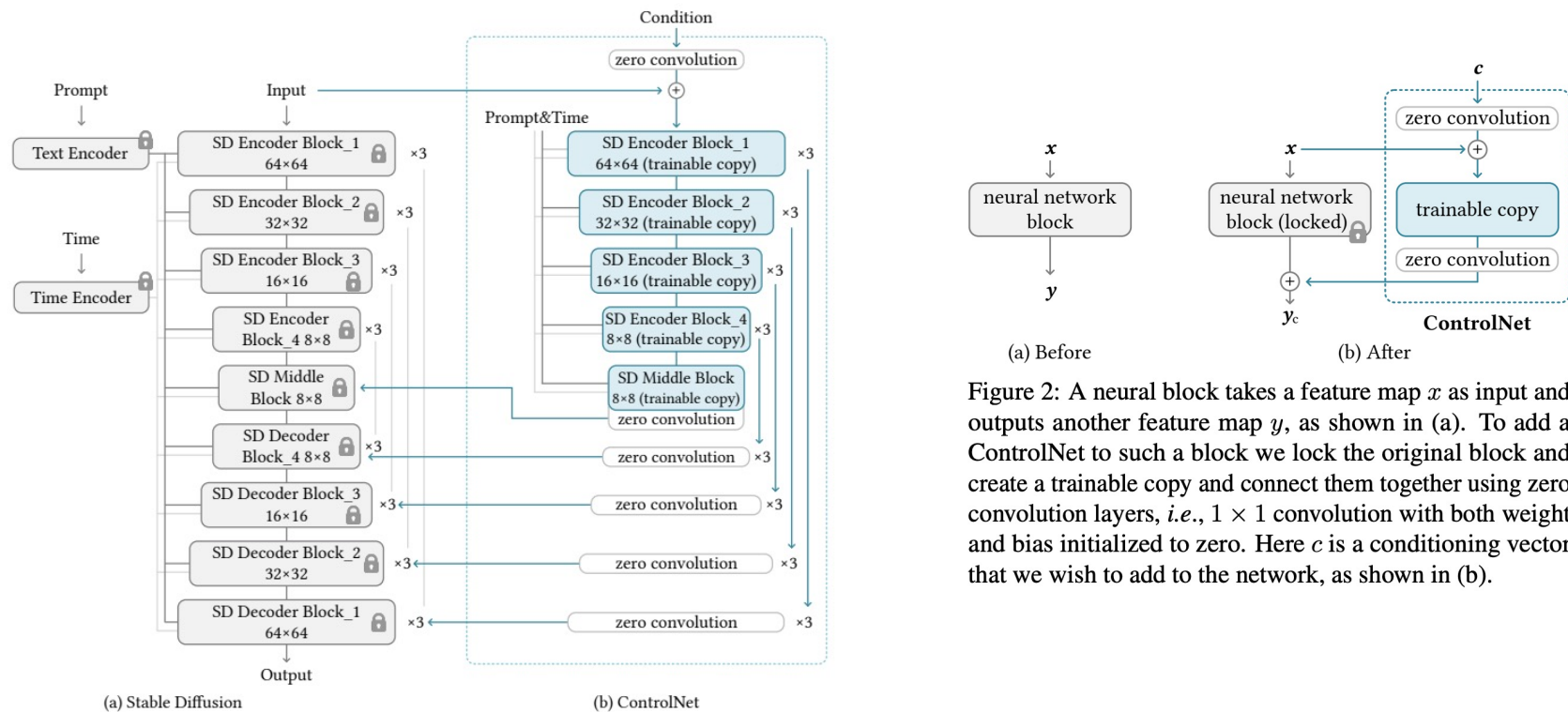
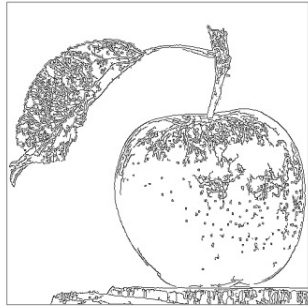


Figure 2: A neural block takes a feature map  $x$  as input and outputs another feature map  $y$ , as shown in (a). To add a ControlNet to such a block we lock the original block and create a trainable copy and connect them together using zero convolution layers, *i.e.*,  $1 \times 1$  convolution with both weight and bias initialized to zero. Here  $c$  is a conditioning vector that we wish to add to the network, as shown in (b).

# ControlNet

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Test input



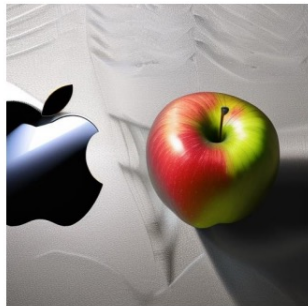
training step 100



step 1000



step 2000



step 6100



**step 6133**



step 8000



step 12000

“The sudden convergence phenomenon. Due to the zero convolutions, ControlNet always predicts high-quality images during the entire training. At a certain step in the training process (e.g., the 6133 steps marked in bold), the model suddenly learns to follow the input condition.”



# ControlNet: Results

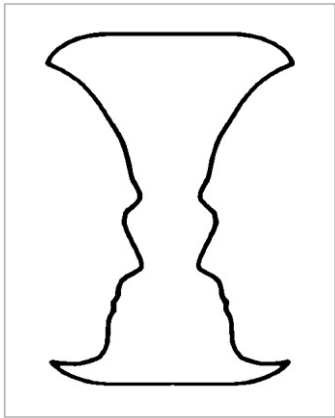


Default prompt: "A high-quality, detailed, and professional image"

Automatic prompt: generate images using default prompt, auto-caption them, use resulting prompt

## ControlNet: Results

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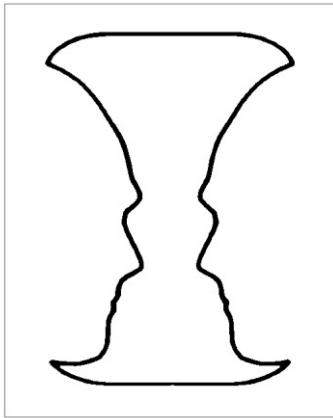
Input



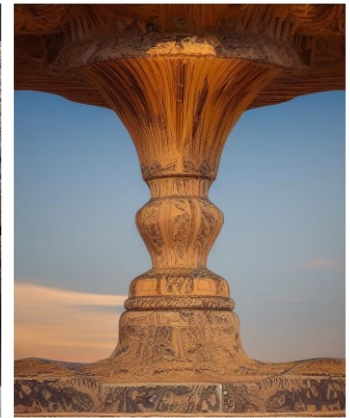
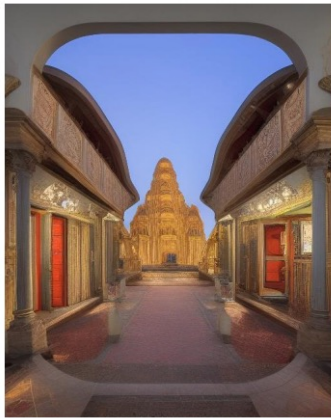
“a high-quality and extremely detailed image”

# ControlNet: Results

---



Input

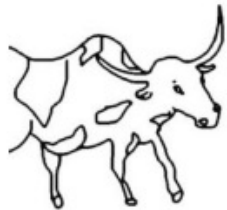
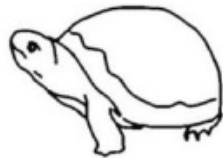


“a high-quality and extremely detailed image”

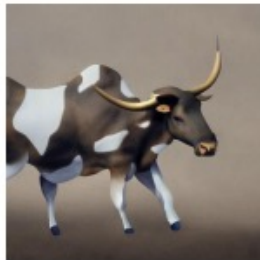
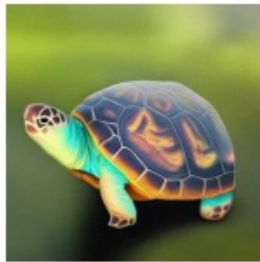


# ControlNet: Results

Input (User Scribble)



Default



Automatic Prompt



"a turtle in river"



"a cow with horns standing in a field"

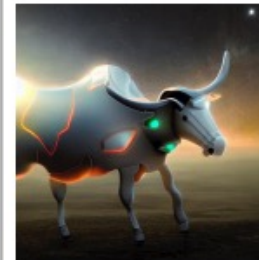


"a digital painting of a hot air balloon"

User Prompt



"a masterpiece of cartoon-style turtle illustration"

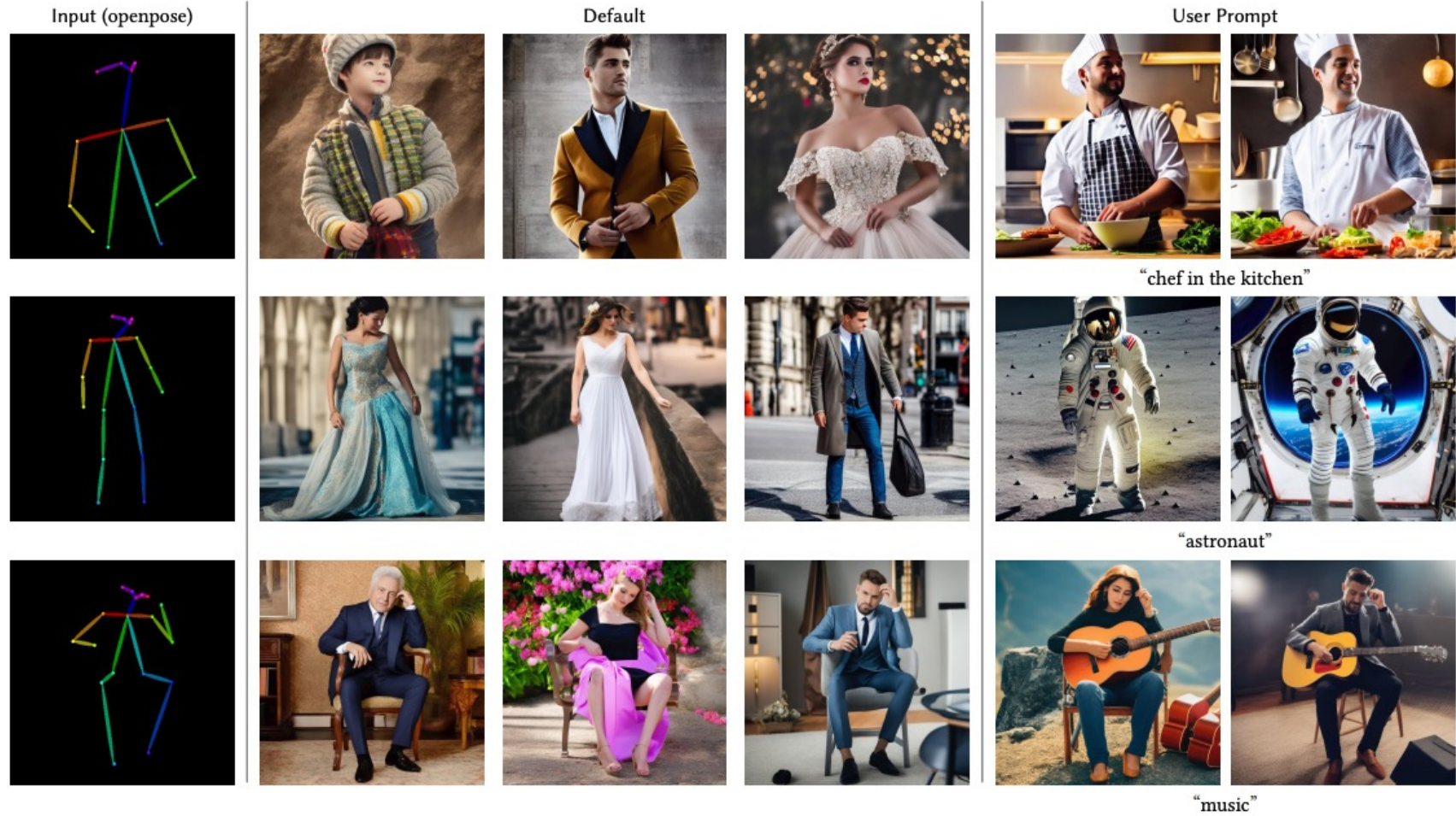


"a robot ox on moon, UE5 rendering, ray tracing"



"magic hot air balloon over a lit magic city at night"

# ControlNet: Results





# ControlNet: Results

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COCO Segmentation



Default

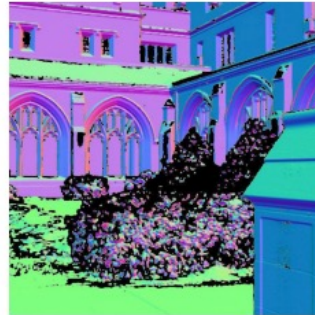


User Prompt



"fantastic artwork, fairy tale"

Normal



Default



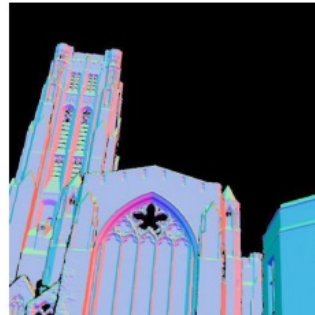
User Prompt



"garden, colorful flowers"



"cyberpunk, city at night"



"Yharnam"



# Outline

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Customizing diffusion models using  
example images

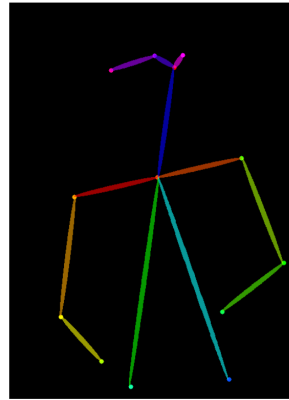


Input images



*in the Acropolis*

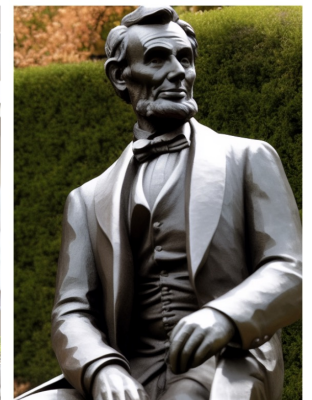
Controlling diffusion models using pixel-aligned conditions



Input human pose



“Chef in kitchen”



“Lincoln statue”

Text-driven image editing



Input



“Make it a Modigliani painting”

# Prompt-to-Prompt Image Editing



"The boulevards are crowded today."



"Children drawing of a castle next to a river."



"Photo of a cat riding on a bicycle."

~~bicycle~~  
car

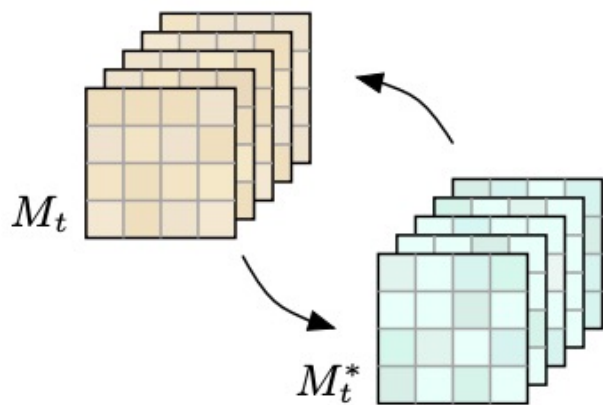
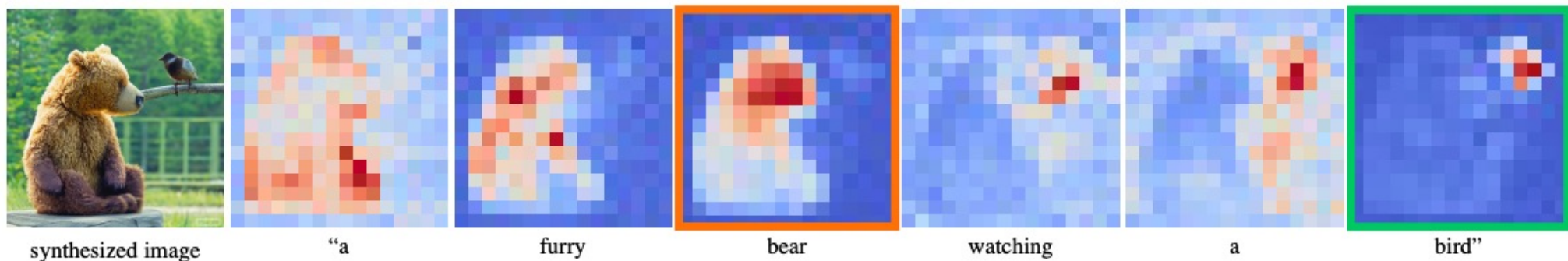


"a cake with decorations."

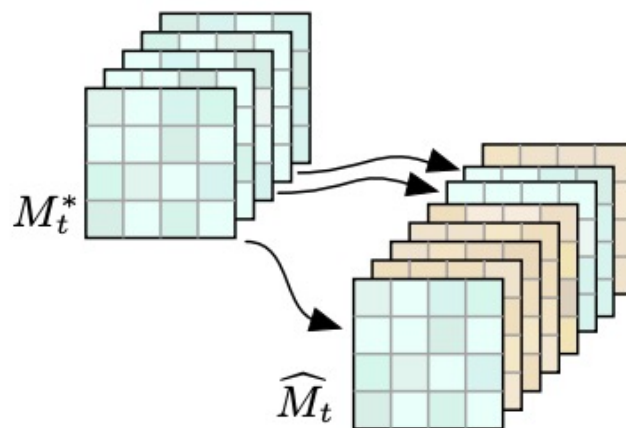
~~decorations~~  
jelly beans

A. Hertz et al. [Prompt-to-Prompt Image Editing with Cross Attention Control](#). ICLR 2023

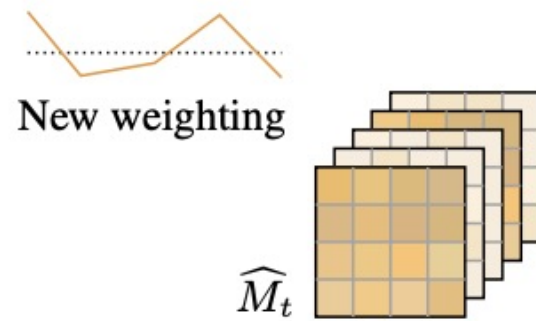
# Prompt-to-Prompt Image Editing



Word Swap



Adding a New Phrase



Attention Re-weighting



# Prompt-to-Prompt Image Editing



Fixed attention maps and random seed

Fixed random seed



# InstructPix2Pix

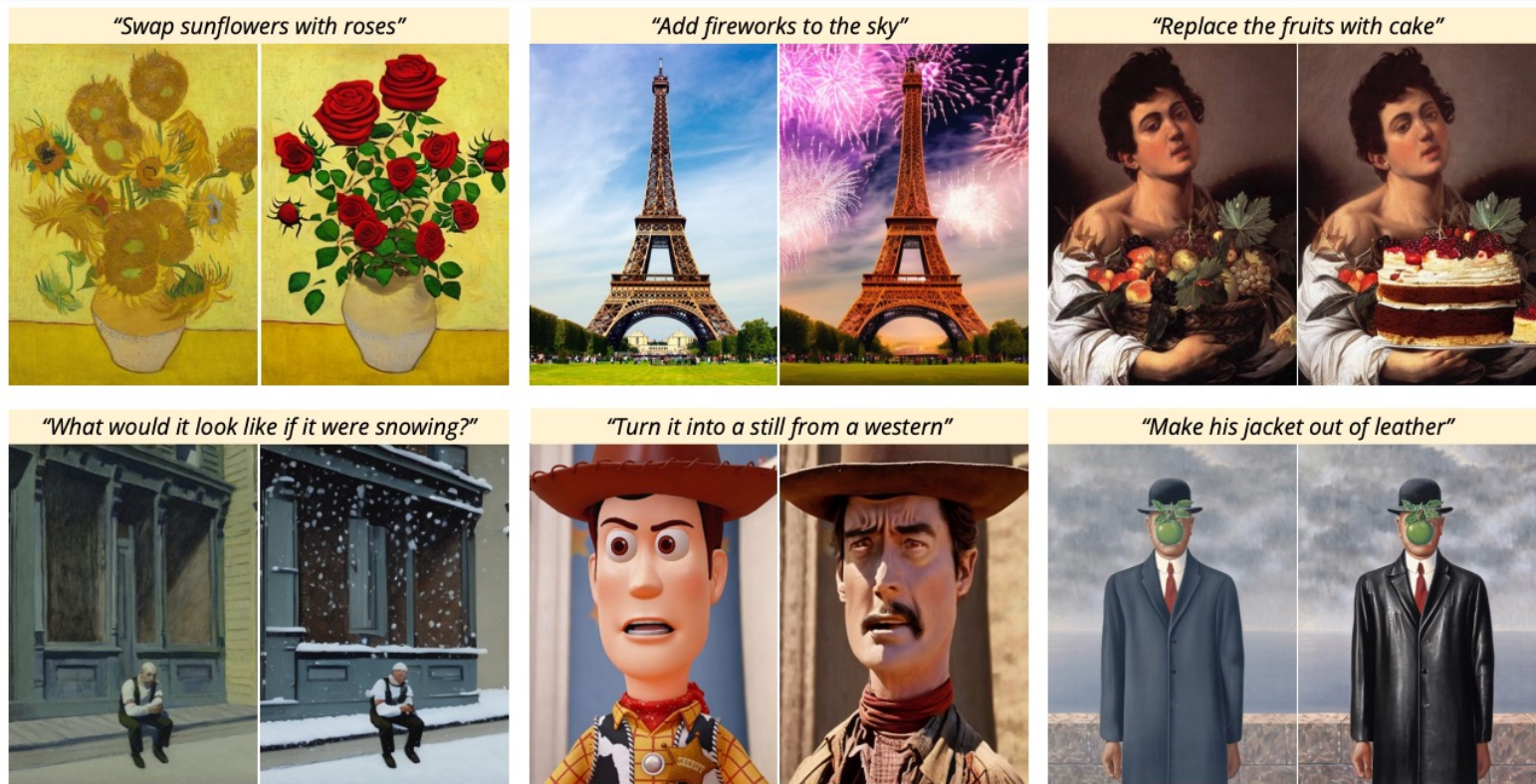


Figure 1. Given **an image** and **an instruction** for how to edit that image, our model performs the appropriate edit. Our model does not require full descriptions for the input or output image, and edits images in the forward pass without per-example inversion or fine-tuning.

T. Brooks, A. Holynski, and A. Efros. [InstructPix2Pix: Learning to Follow Image Editing Instructions](#). CVPR 2023



# InstructPix2Pix

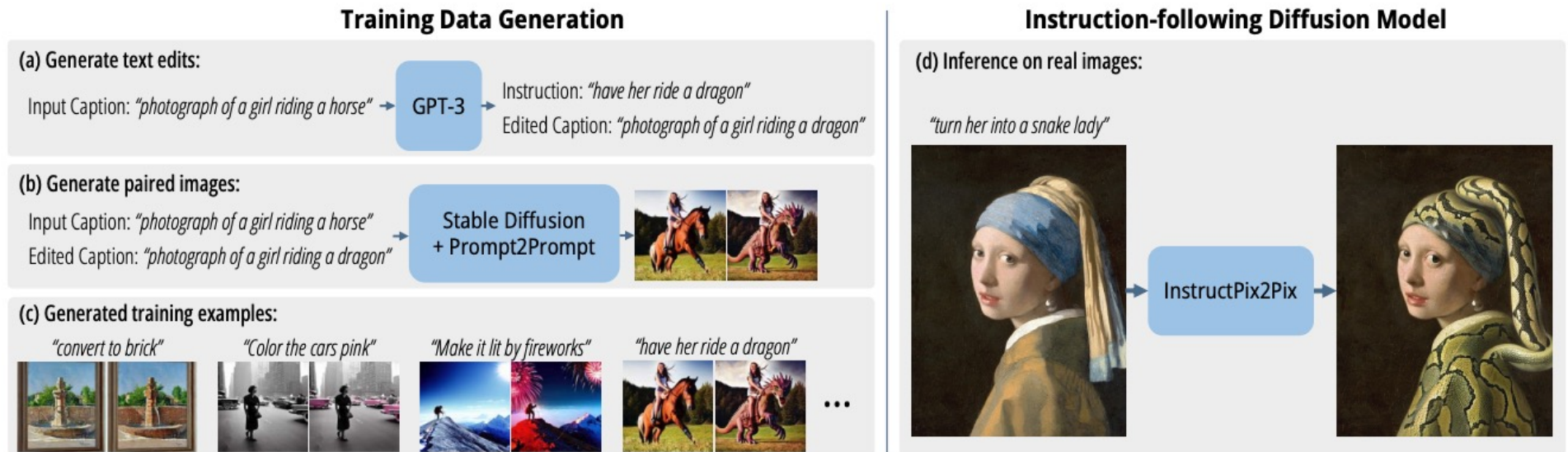


Figure 2. Our method consists of two parts: generating an image editing dataset, and training a diffusion model on that dataset. (a) We first use a finetuned GPT-3 to generate instructions and edited captions. (b) We then use StableDiffusion [52] in combination with Prompt-to-Prompt [17] to generate pairs of images from pairs of captions. We use this procedure to create a dataset (c) of over 450,000 training examples. (d) Finally, our InstructPix2Pix diffusion model is trained on our generated data to edit images from instructions. At inference time, our model generalizes to edit real images from human-written instructions.

T. Brooks, A. Holynski, and A. Efros. [InstructPix2Pix: Learning to Follow Image Editing Instructions](#). CVPR 2023



# InstructPix2Pix

- Fine-tuning GPT-3:

|                                            | Input LAION caption                                                                                   | Edit instruction                       | Edited caption                                                                                                                                |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Human-written</b><br>(700 edits)        | <i>Yefim Volkov, Misty Morning</i>                                                                    | <i>make it afternoon</i>               | <i>Yefim Volkov, Misty Afternoon</i>                                                                                                          |
|                                            | <i>girl with horse at sunset</i>                                                                      | <i>change the background to a city</i> | <i>girl with horse at sunset in front of city</i>                                                                                             |
|                                            | <i>painting-of-forest-and-pond</i>                                                                    | <i>Without the water.</i>              | <i>painting-of-forest</i>                                                                                                                     |
|                                            | ...                                                                                                   | ...                                    | ...                                                                                                                                           |
| <b>GPT-3 generated</b><br>(>450,000 edits) | <i>Alex Hill, Original oil painting on canvas, Moonlight Bay</i>                                      | <i>in the style of a coloring book</i> | <i>Alex Hill, Original coloring book illustration, Moonlight Bay</i>                                                                          |
|                                            | <i>The great elf city of Rivendell, sitting atop a waterfall as cascades of water spill around it</i> | <i>Add a giant red dragon</i>          | <i>The great elf city of Rivendell, sitting atop a waterfall as cascades of water spill around it with a giant red dragon flying overhead</i> |
|                                            | <i>Kate Hudson arriving at the Golden Globes 2015</i>                                                 | <i>make her look like a zombie</i>     | <i>Zombie Kate Hudson arriving at the Golden Globes 2015</i>                                                                                  |
|                                            | ...                                                                                                   | ...                                    | ...                                                                                                                                           |

Table 1. We label a small text dataset, finetune GPT-3, and use that finetuned model to generate a large dataset of text triplets. As the input caption for both the labeled and generated examples, we use real image captions from LAION. Highlighted text is generated by GPT-3.

# InstructPix2Pix

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- Generating input-output image pairs:



(a) Without Prompt-to-Prompt.

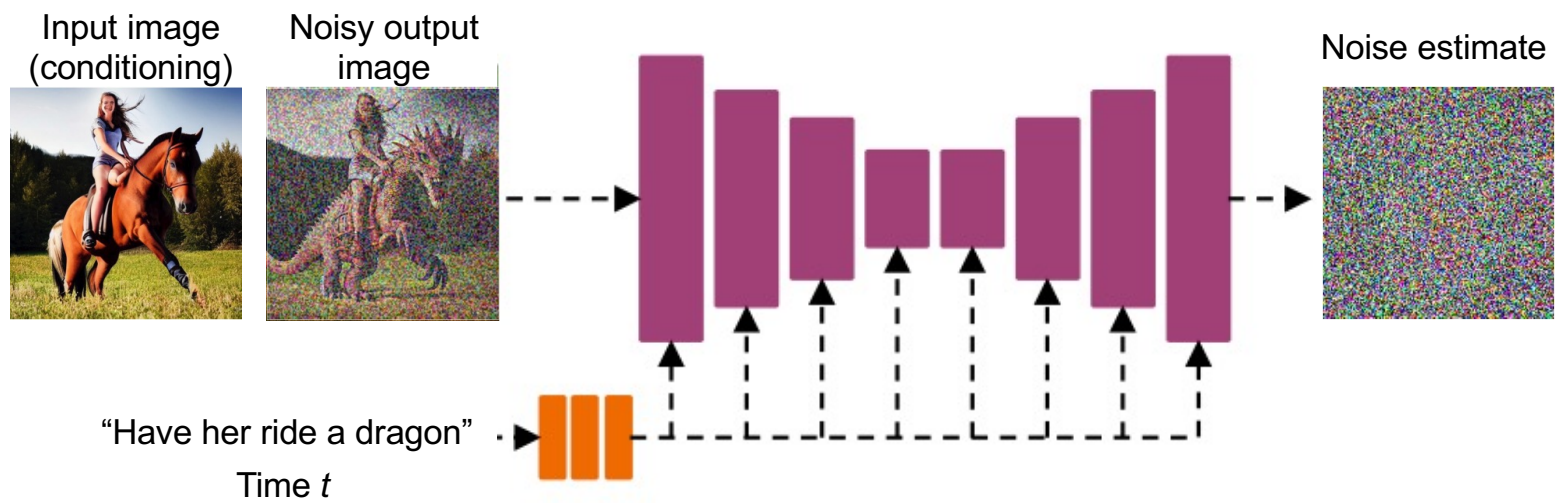
(b) With Prompt-to-Prompt.

Figure 3. Pair of images generated using StableDiffusion [52] with and without Prompt-to-Prompt [17]. For both, the corresponding captions are “*photograph of a girl riding a horse*” and “*photograph of a girl riding a dragon*”.

# InstructPix2Pix

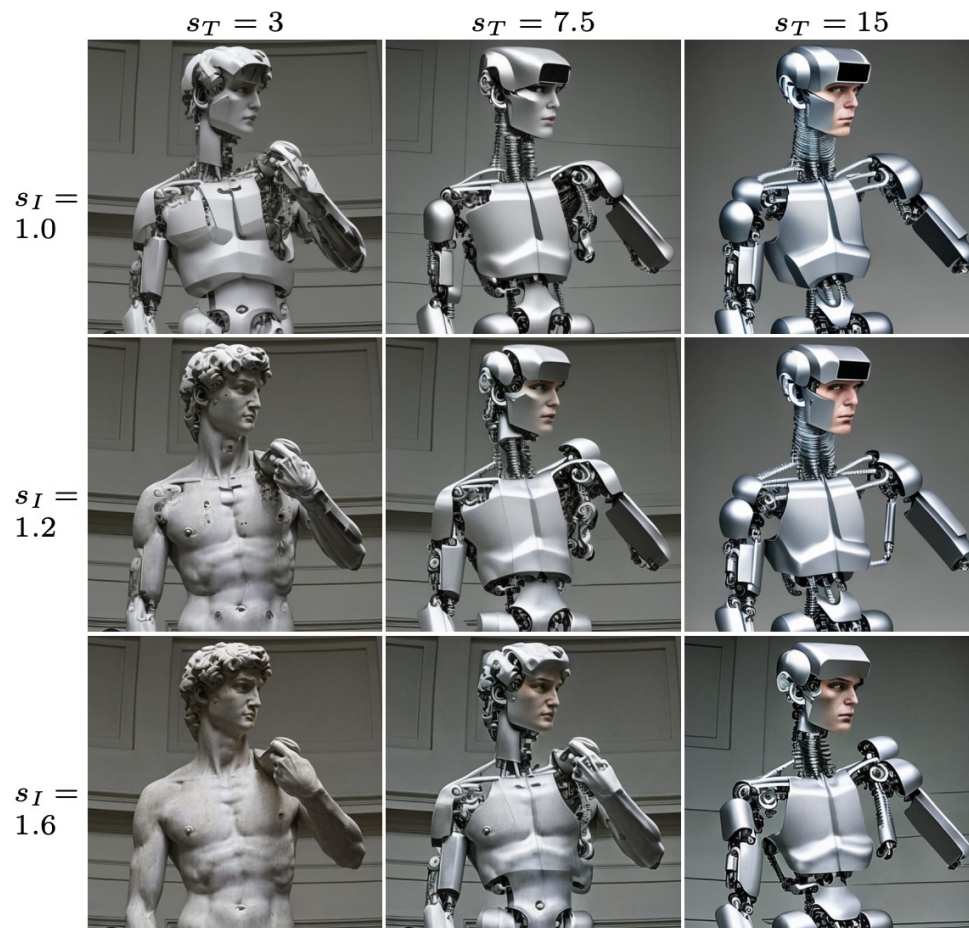
---

- Fine-tuning a DM for image-to-image translation:





# InstructPix2Pix



Edit instruction: "Turn him into a cyborg!"

Figure 4. Classifier-free guidance weights over two conditional inputs.  $s_I$  controls similarity with the input image, while  $s_T$  controls consistency with the edit instruction.

# InstructPix2Pix: Results

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Figure 5. *Mona Lisa* transformed into various artistic mediums.



Figure 6. *The Creation of Adam* with new context and subjects (generated at 768 resolution).



# InstructPix2Pix: Results

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Input



*"Apply face paint"*



*"What would she look like as a bearded man?"*



*"Put on a pair of sunglasses"*



*"She should look 100 years old"*



*"What if she were in an anime?"*



*"Make her terrifying"*



*"Make her more sad"*



*"Make her James Bond"*



*"Turn her into Dwayne The Rock Johnson"*



# InstructPix2Pix: Results

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*"Make it Paris"*



*"Make it Hong Kong"*



*"Make it Manhattan"*



*"Make it Prague"*



*"Make it evening"*



*"Put them on roller skates"*



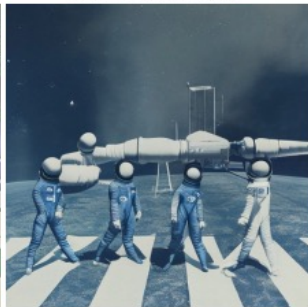
*"Turn this into 1900s"*



*"Make it underwater"*



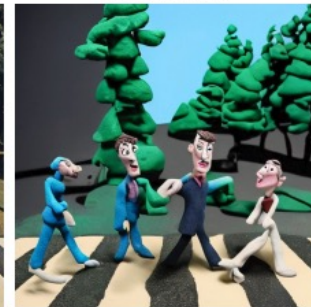
*"Make it Minecraft"*



*"Turn this into the space age"*



*"Make them into Alexander Calder sculptures"*



*"Make it a Claymation"*

# InstructPix2Pix: Failure cases

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*"Zoom into the image"*



*"Move it to Mars"*



*"Color the tie blue"*



*"Have the people swap places"*

Figure 13. Failure cases. Left to right: our model is not capable of performing viewpoint changes, can make undesired excessive changes to the image, can sometimes fail to isolate the specified object, and has difficulty reorganizing or swapping objects with each other.